

Single Collective: Heuristic vs. MILP

Exact validation with absolute Avg JCT, makespan, and solver runtime.

Tenants	Heur. Avg JCT (s)	ILP Avg JCT (s)	Avg JCT Gap (%)	Heur. Makespan (s)	ILP Makespan (s)	Makespan Gap (%)	Heur. Runtime (s)	ILP Runtime (s)
2	0.013432	0.013432	0.0000	0.013432	0.013432	0.0000	0.382	312.8
3	0.013431	0.013429	0.0130	0.016111	0.016111	0.0000	0.029	128.0
4	0.006714	0.006714	0.0000	0.010743	0.010743	0.0000	0.005	3.0

Absolute values; no normalization. Gap = (Heuristic - ILP) / ILP.